

Assignment 1

1

We have seen that the same dataset may be modeled differently depending on the application context, and that modern data systems need to deal with the challenges of Volume, Velocity, Variety, and Veracity (the “4Vs” as data challenges). For each of the following datasets:

1. Identify applicable data types, choosing from:
 - structured / semi-structured / unstructured
 - ordered data / time-series data
2. Identify one or two dominant “V(s)” among the 4Vs that you believe would be the major data challenge(s) to deal with.
3. Briefly explain your choices in (a) and (b). A dataset may belong to multiple categories, depending on the context.

a

Smart home IoT data collected from thousands of households, including temperature, motion sensors, and power usage, streamed every second.

✓ Answer ✓

Structured, time-series data

Structured as we have pre-defined data collection types across all houses in a standard synchronized time frame (every second)

Likely Volume and Velocity

Since we will be getting a large amount of data every second, dealing with the volume and velocity will be a difficult task.

b

A large-scale social network dataset capturing users, friendships, and interactions (likes, comments, shares), used for community detection and recommendation.

✓ Answer

Possibly semi-structured, ordered

We don't know if the data itself, especially interactions is well structured, as there are many possible kinds of interactions. Since we are not recording data on a specific period of time, we may or may not have complete data, instead possibly opting for samples of the dataset, possibly without timestamps.

Volume, Variety

Since the data could come in many ways, variety may make it difficult to analyse as well as the large amount of events happening on a social network

c

Logs of user interactions on an online learning platform, recording page views, quiz attempts, and timestamps for millions of students.

✓ **Answer**

Structured, time-series

Structured as there is a well defined format of exactly what data is available (page views, quizzes, etc) and time-series as we have full collection across the entire service along with timestamps.

Volume and Velocity likely

Since we are servicing millions of students, we likely will also be processing large volumes of data quickly.

d

A corpus of research papers (PDFs) used to build a semantic search engine for scientific literature.

✓ **Answer**

Unstructured, ordered

As we do not know what format the data within the PDFs are in, whether there be text or images or graphs etc. Ordered since we do not care or know the timestamps of the entire sample of data, while individual data points have ordered sequential data within them

Variety and Volume

We likely will be dealing with a huge amount of very varied PDFs, and processing both the amount of PDFs and extracting useful information from an unstructured varied format may prove difficult.

e

Autonomous driving data consisting of synchronized camera images, LiDAR scans, and GPS signals collected during road tests.

✓ **Answer**

Semi-structured data, time-series

Since each individual source may be structured, while their relation to each other is not relational nor modelled, this would be semi-structured. Since it is in order sampled with timestamps, this is likely a time-series

Volume and Variety

Sensor fusion is a notoriously hard task and especially for a autonomous car with a large volume of information, this may be an even more difficult task.

f

A dataset of extracted features from medical images, represented as records with attributes {patient id, image id, feature vector, diagnosis label}.

✓ Answer

Structured, ordered

Since we do not necessarily know or care about the time of the data this would be ordered. Since we already have all pieces of data relationally, this is structured.

Veracity and Volume

Since data is critical to be verified and accurate, veracity is of utmost importance. Volume is also a concern as there are likely a huge amount of medical records to work through.

g

Product listings from multiple e-commerce websites, collected via web scraping and stored in JSON format.

✓ Answer

semi-structured, ordered

Since we are using a JSON format with no specific schema or relational design, this would be considered semi-structured. Our data is also not necessarily timed or time based either, leading this to be an ordered set.

Variety and veracity

Since there is going to be a lot of different formats for websites, dealing with this variance in a meaningful way will be difficult. Veracity is also important as your dataset is no verified in any way and you may get many false records of things from things like scraping.

h

Real-time clickstream data from a large online advertising platform used to update bidding strategies.

✓ Answer

structured, time-series

Since this likely relational, it would be structured, and since it is live data it is also likely time-series.

Velocity and Volume

Since we are once again dealing with a lot of data and very quickly, volume and velocity become an immediate concern.

2

Consider the following schema for FreshDash, an online food delivery and recommendation platform. It keeps track of restaurants, customers, menu items, orders, deliveries, and customer reviews. An order represents a purchase placed by a customer from a restaurant at a specific time.

Schema (6 relations)

```
Restaurant(rest_id, rest_name, city, street_address, cuisine_type)
MenuItem(item_id, rest_id, item_name, category, price)
Customer(cust_id, cust_name, email, phone, home_zip)
Order(order_id, cust_id, rest_id, order_time, total_amount)
OrderItem(order_id, item_id, quantity, item_price_at_order_time)
Review(review_id, order_id, cust_id, rating, review_time)
```

The following facts have been validated.

- Each restaurant has a unique rest id; the combination (rest name, street address, city) is also unique.
- A restaurant can offer many menu items; each menu item belongs to exactly one restaurant.
- Each customer has a unique cust id and a unique email.
- A customer can place many orders; each order is placed by exactly one customer.
- Each order is associated with exactly one restaurant.
- An order can include multiple menu items, and a menu item can appear in many orders.
- For each (order id, item id) pair, there is at most one OrderItem record.
- A customer may write at most one review per order.
- Ratings are integers in {1, 2, 3, 4, 5}.

a

(10 points) Identify a primary key and a candidate key for each relation.

✓ Answer

Restaurant: PK rest_id, CK (rest_name, street_address, city)

MenuItem: PK item_id, CK - we cannot assume that all items have unique names

Customer: PK cust_id, CK (email)

Order: PK order_id, no CK

OrderItem: PK order_id, item_id, no CK

Review: PK review_id, CK (order_id)

b

(10 points) Identify the foreign keys of each relation, given your choice of primary keys.

✓ **Answer**

MenuItem(rest_id) -> Restaurant(rest_id)

Order(cust_id) -> Customer(cust_id)

Order(rest_id) -> Restaurant(rest_id)

OrderItem(order_id) -> Order(order_id)

OrderItem(item_id) -> MenuItem(item_id)

Review(order_id) -> Order(order_id)

Review(cust_id) -> Customer(cust_id)

c

(10 points) Identify the foreign keys that are also a part of the primary keys of the same relation they are defined on.

✓ **Answer**

Just OrderItem(order_id, item_id)

3

Continue with the above schema. FreshDash wants to support personalized food recommendations. To do this, the system must track:

- which customers ordered which menu items,
- from which restaurants,
- and at what time.

a

(10 points) Based on the existing schema: identify the relation(s) needed to answer these questions.

✓ **Answer**

Customer, Order, OrderItem, MenuItem, Restaurant

b

(10 points) As a data model designer, you need to monitor potential “harmful” data modifications that may cause violations. Choose any three data constraints on the FreshDash schema, and for each data constraint, give one possible modification that may cause the violation of the constraint. Briefly explain — you can give an example with some tuples.

✓ Answer

Inserting a new user with a preexisting email in the Customer table

Multiple reviews inserted for the same order_id

A rating gets inserted that is not within 1..5.

4

- Build your own toy structured dataset with the “Healthsense” toy dataset, provided as a running demo in Module 2.
- Set up a Github repo to manage your R&D project and relevant code for CSDS 234.

For your answer Submit the following (either hosted in your GitHub project, or uploaded in a zip folder):

- One SQL file, named first name 234 A1.sql. This file must include:
 - CREATE TABLE statements
 - Primary keys
 - At least one foreign key
 - At least one constraint (CHECK or NOT NULL)
 - INSERT statements (3–5 rows per table)
- Supplement materials with prefix “first name 234 A1”:
 - run “Select * from ...” statements, to show that you have successfully loaded the csv to the structured relations. Take screenshots of the results and submit them.
 - (Optional, no need to submit): try some “harmful” insertion that may violate the integrity constraints. What will PostgreSQL report? Observe and think about why.

```
postgres=# SELECT * FROM diagnoses;
```

diag_id	visit_id	code	description
1001	101	R53	Fatigue
1002	101	R51	Headache
1003	102	I10	Hypertension
1004	103	E11	Type 2 diabetes
1005	104	J06	Common cold
1006	105	E66	Obesity
1007	106	G47	Sleep disorder
1008	107	R07	Chest pain
1009	108	I10	Hypertension
1010	110	I10	Hypertension
1011	111	R53	Fatigue
1012	112	F41	Anxiety

(12 rows)

And when I try to insert NULLs:

```
postgres=# INSERT INTO users (age) VALUES (10);  
ERROR:  null value in column "name" of relation "users" violates not-null  
constraint  
DETAIL:  Failing row contains (1, null, 10, null, null).
```

This is because I did not fill in all the values in the row for all the other columns other than age.

The rest of the sql code along with devshells can be found on:

<https://github.com/tnichols217/postgres-practice>